

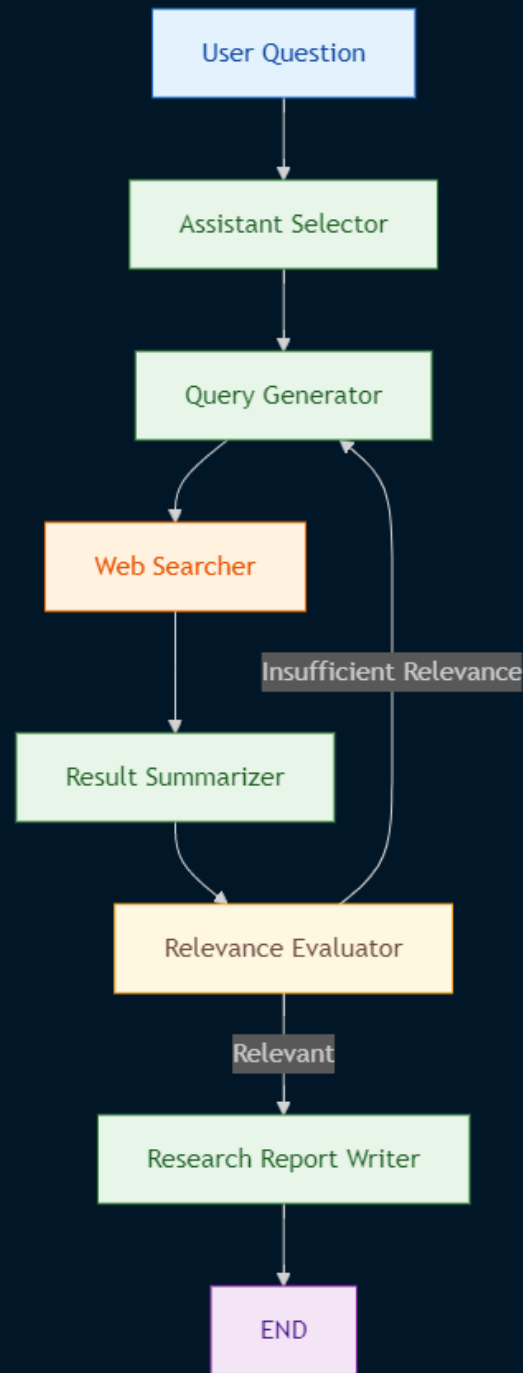
Agentic AI Engineering

***Agentic Research Workflow
with
LangGraph and Adaptive Query Regeneration***

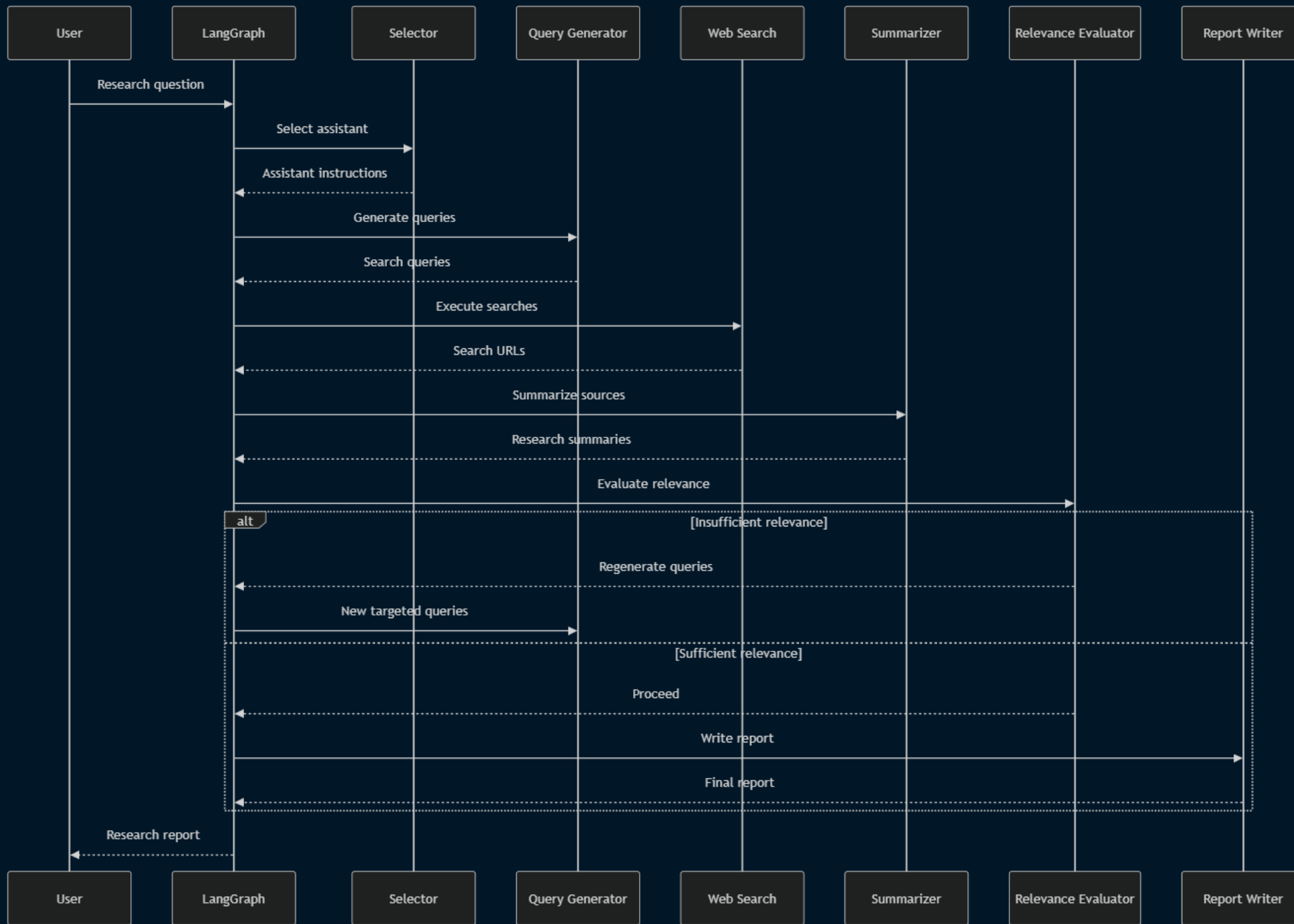
Max Montaseri

- Designed a **LangGraph StateGraph agentic workflow** with typed state management and six specialized research nodes spanning assistant selection, query generation, web retrieval, summarization, relevance evaluation, and report generation.
- Implemented **conditional graph routing** that evaluates retrieved evidence and dynamically loops back to query generation when search relevance falls below a defined threshold.
- Built adaptive multi-query research strategies that generate increasingly targeted search queries across successive retrieval iterations while enforcing a maximum workflow iteration count.
- Developed resilient web-search orchestration with **rate limiting, retry logic, exponential backoff, jitter** and **fallback retrieval mechanisms**.
- Implemented source-aware document summarization using web scraping, LLM summarization, provenance metadata, and explicit fallback-source annotations.
- Designed a typed ResearchState using TypedDict structures to maintain query, retrieval, summary, evaluation, and final-report state across LangGraph nodes.
- Externalized assistant-selection, search-query, summarization, and report-generation prompts to support modular prompt engineering and independent prompt iteration.
- Built defensive parsing and failure-recovery paths for malformed LLM JSON, failed searches, unavailable webpages, insufficient content, and failed relevance evaluation.
- Established an extensible foundation for enterprise RAG, vector retrieval, reranking, source verification, citation tracking, and multi-agent supervisor architectures.

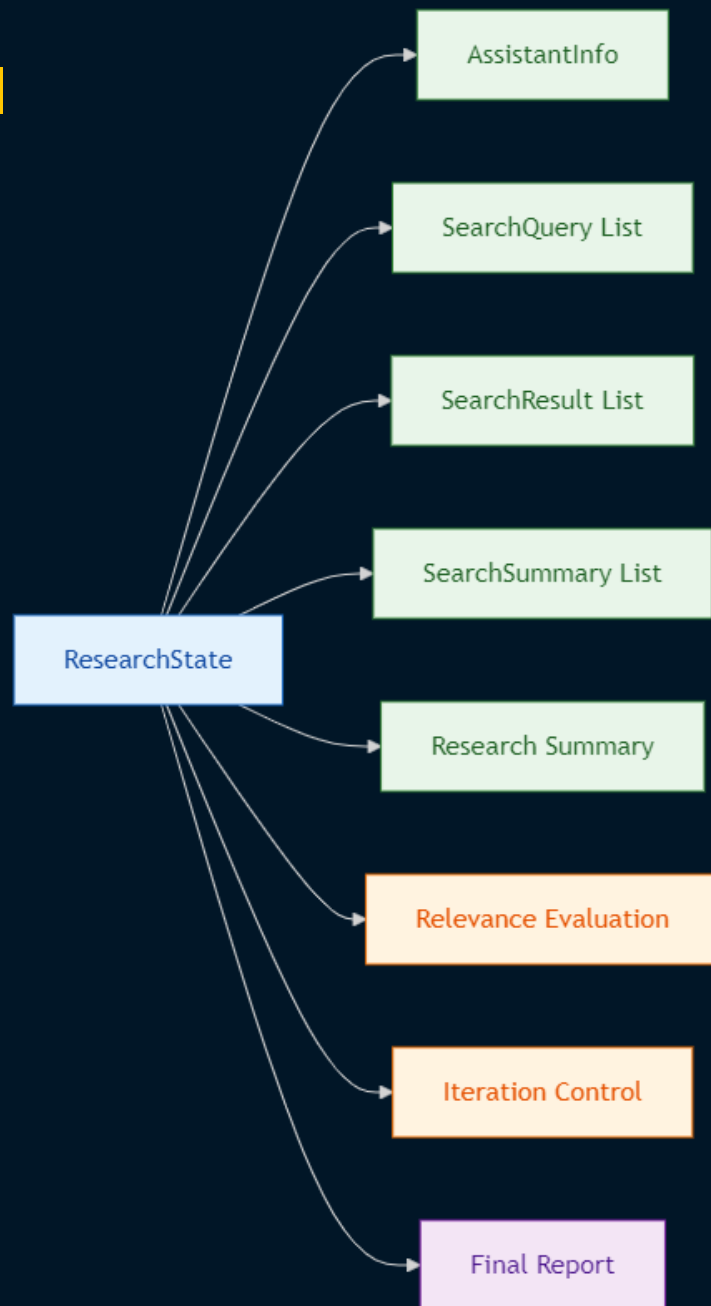
Architecture



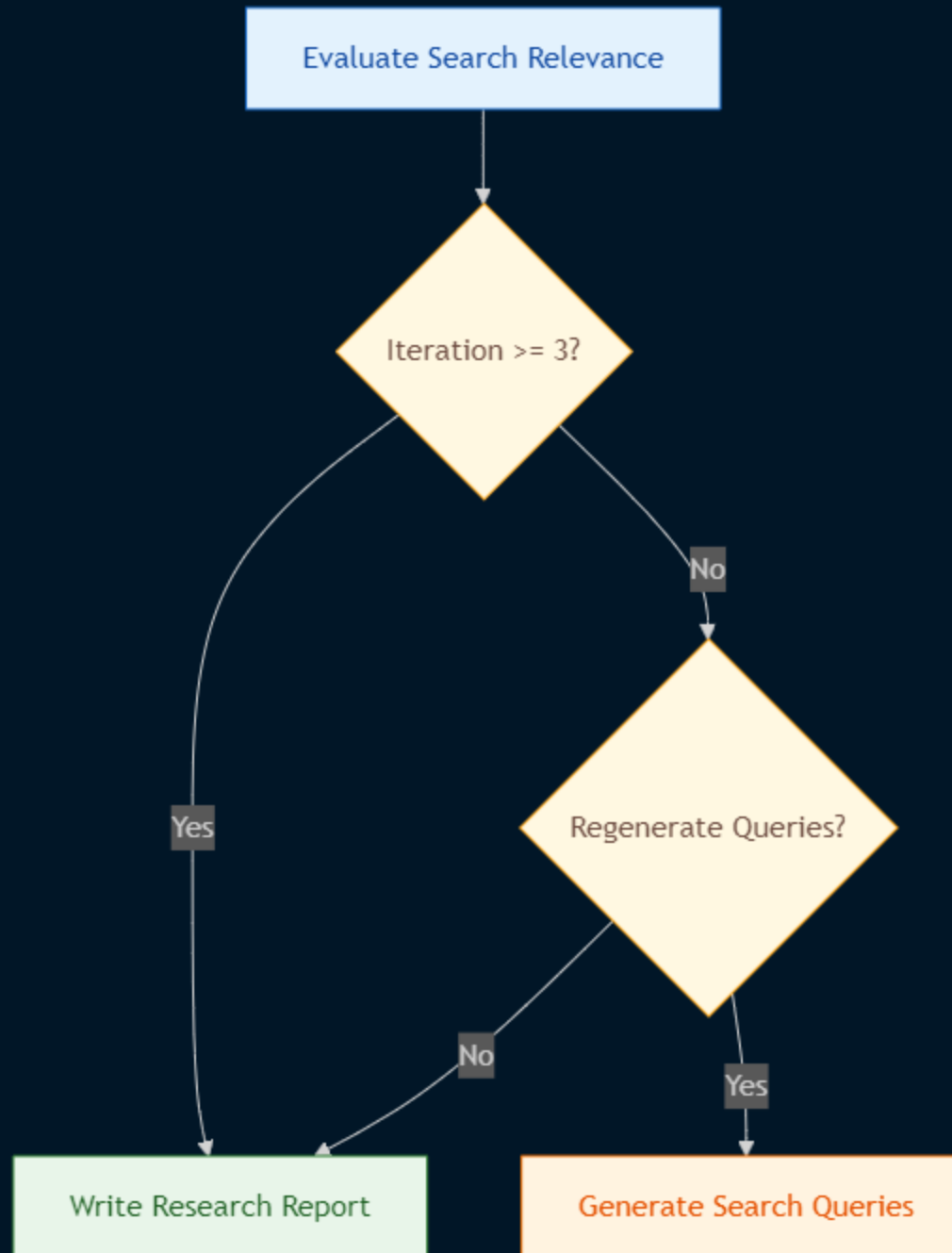
Agentic Workflow



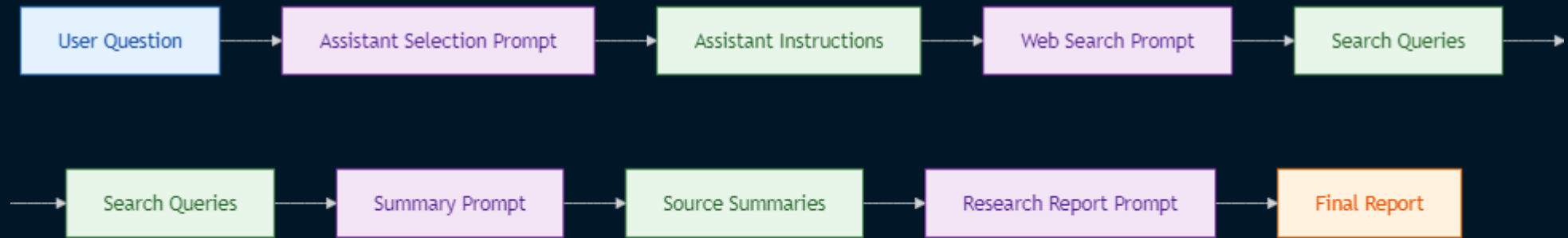
LangGraph State Model



Conditional Routing and Iteration



Prompt architecture



Production Enhancement

Advanced Agentic Routing

